Margarita Grushanina

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Employment History

2024 – present — **Research Associate for Bayesian Statistics and Causal Inference**, Department of Epidemiology and Biostatistics, School of Public Health, Imperial College London (London, United Kingdom).

2014 – 2024 — **Economist/Quantitative Analyst**, Erste Group Bank AG, Research Department (Vienna, Austria).

2011 – 2013 — **Bank Analyst**, Erste Group Bank AG, International Business Department (Vienna, Austria).

2007 – 2011 — **Data Analyst**, Erste Group Bank AG, Strategic Risk Management Department (Vienna, Austria).

2006 — **Researcher**, United Nations Industrial Development Organisation (UNIDO), Strategic Research and Economics Branch (Vienna, Austria).

2005 — **Intern**, United Nations Industrial Development Organisation (UNIDO), Strategic Research and Economics Branch (Vienna, Austria).

2004 – 2005 — **Lecturer**, Saint Petersburg State University, Department of Economics (Saint Petersburg, Russia).

Education

2018 – 2023 — **Doctoral programme**, Vienna University of Economics and Business. Institute of Quantitative Economics, Institute of Mathematics and Statistics (Vienna, Austria).
Advisors: Sylvia Frühwirth-Schnatter and Alfred Stiassny
Dissertation: *Bayesian Methods for Unsupervised Data Analysis in Application to Data Sets Exhibiting Non-Gaussianity*.

2001 – 2004 — **Candidate of Science** in Mathematical Methods in Economics, Saint Petersburg State University (Saint Petersburg, Russia). Russian postgraduate research degree.
Advisor: Galina Chernova

1996 – 2001 — **Diploma (with honours)** in Mathematical Methods in Economics, Saint Petersburg State University (Saint Petersburg, Russia).
Advisor: Andrey Kudryavtsev

Schools and Training

Jun 2023 — **Workshop "Causal Inference: Concepts, Tools and Practice"**, Vienna University of Economics and Business (Vienna, Austria).
Instructors: Susanne Strohmaier, Michael Kammer (Medical University of Vienna)

Nov 2021 — **Tutor-Led Hybrid Training Course "Machine Learning Training for Industry"**, Mind Project and Newton Gateway to Mathematics, University of Cambridge (Cambridge, UK).
Instructor: Simon Walkowiak (UCL and Mind Project)

Schools and Training (continued)

Aug 2018 — **Summer School "Uncertainty Quantification for PDEs: Numerical Analysis and Scientific Computing"**, ETH Zürich (Zürich, Switzerland).
Instructors: A. Barth (Uni Stuttgart), A. Cohen, G. Migliorati (Sorbonne), H. Harbrecht (Uni Basel), C. Jerez-Hanckes (PUC Santiago), S. Mishra (ETH Zürich), C. Schwab (ETH Zürich), B. Sudret (ETH Zürich).

Jul 2018 — **Summer School "Machine Learning Methods and Data Analytics in Risk and Insurance"**, Vienna University of Technology (Vienna, Austria).
Instructors: Gareth Peters (Heriot-Watt University), Pavel Shevchenko (Macquarie University).

Publications

Journal articles

1. Grushanina, M., & Frühwirth-Schnatter, S. (2025). Dynamic Mixture of Finite Mixtures of Factor Analysers. *Bayesian Analysis. Advance publication*, 1–30. doi:10.1214/25-BA1543

Preprints

1. Grushanina, M. (2026). Weak Factor Recovery in Sparse Bayesian Factor Models under Exchangeable Shrinkage Process Prior. arXiv: 2610.40491
2. Grushanina, M. (2023). A Review of Bayesian Methods for Infinite Factorisations. arXiv: 2309.12990

Conference proceedings

1. Grushanina, M., & Frühwirth-Schnatter, S. (2021). Bayesian infinite factor models with non-Gaussian factors. In *JSM Proceedings, International Society for Bayesian Analysis (ISBA) Section* (pp. 396–415). Alexandria, VA: American Statistical Association. Retrieved from https://ww2.amstat.org/meetings/proceedings/2021/data/assets/pdf/1913713.pdf

Discussions

1. Grushanina, M. (2025). Discussion on Sparse Bayesian Factor Analysis When the Number of Factors Is Unknown by S. Frühwirth-Schnatter, D. Hosszejni, and H. F. Lopes. *Bayesian Analysis, 20(1)*, pp. 316–318. doi:10.1214/24-BA1423

Presentations, Conferences and Seminars

Invited presentations

Dec 2024 — *Dynamic Mixture of Finite Mixtures of Factor Analysers*, 18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics) 2024 (London, UK).

Oct 2023 — *Dynamic Mixture of Finite Mixtures of Factor Analysers with Automatic Inference on the Number of Clusters and Factors*, International Day of Women in Statistics and Data Science (IDWSDS) 2023 (virtual).

Dec 2022 — *Dynamic Mixture of Finite Mixtures of Factor Analysers with Automatic Inference on the Number of Clusters and Factors*, 15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics) 2022 (London, UK).

Presentations, Conferences and Seminars (continued)

Contributed presentations

Sept 2026 — *Multi-Subject Factor Regression Biclustering for scRNA-seq Data*, MASAMB 2026 (Cambridge, UK).

Apr 2026 — *Multi-Study Factor Regression Biclustering*, Research Computing Showcase Day 2026 (London, UK).

Dec 2023 — *Dynamic Mixture of Finite Mixtures of Factor Analysers with Automatic Inference on the Number of Clusters and Factors*, European Winter Meeting of The Econometric Society (EWMES) 2023 (Manchester, UK).

Sep 2023 — *Dynamic Mixture of Finite Mixtures of Factor Analysers with Automatic Inference on the Number of Clusters and Factors*, European Seminar on Bayesian Econometrics (ESOBE) 2023 (Glasgow, UK).

Sep 2022 — *Dynamic Mixture of Finite Mixtures of Factor Analysers with Automatic Inference on the Number of Clusters and Factors*, European Seminar on Bayesian Econometrics (ESOBE) 2022, Young Researcher Session (Salzburg, Austria).

Apr 2022 — *Mixture of Factor Analysers and Adaptive Telescoping Sampler for Automatic Inference on the Number of Clusters and Factors*, BNP 2022 Networking Workshop (Nicosia, Cyprus).

Aug 2021 — *Bayesian Infinite Factor Models with Non-Gaussian Factors*, Joint Statistical Meetings 2021 (virtual).

Jul 2021 — *Bayesian Infinite Factor Models with Non-Gaussian Factors*, 2021 ISBA World Meeting (virtual).

Posters

Apr 2026 — *Multi-Study Factor Regression Biclustering*, London Meeting on Computational Statistics (London, UK).

Jul 2024 — *Dynamic Mixture of Finite Mixtures of Factor Analysers*, 2024 ISBA World Meeting (Venice, Italy).

Jun 2022 — *Mixture of Factor Analysers and Adaptive Telescoping Sampler for Automatic Inference on the Number of Clusters and Factors*, 2022 ISBA World Meeting (Montreal, Canada).

Jun 2021 — *Bayesian Infinite Factor Models with Non-Gaussian Factors*, 2021 ISBA World Meeting (virtual).

Research seminars and workshops

May 2026 — *Multi-Study Factor Regression Biclustering*, 4th Vienna Interinstitutional Bayesfest, Vienna University of Technology (Vienna, Austria).

Nov 2025 — *Multi-Study Factor Regression Biclustering*, 3rd joint workshop TU/WU/ARI, Austrian Academy of Sciences (Vienna, Austria).

Jan 2025 — *Dynamic Mixture of Finite Mixtures of Factor Analysers*, European University at St. Petersburg (EUSP) Research Seminar (virtual).

Dec 2021 — *Bayesian Infinite Factor Models with Non-Gaussian Factors*, Center for Econometrics and Business Analytics (CEBA) Research Seminar (virtual).

Teaching experience

Autumn term 2026	*Teaching Assistant*, Introduction to Statistical Thinking and Data Analysis, MSc Epidemiology and MSc Health Data Analytics and Machine Learning, Imperial College London.
Spring term 2026	*Lecturer*, Bayesian Reasoning and Methods for Spatio-Temporal Data, MSc Epidemiology, Imperial College London.
Winter term 2004/2005	*Lecturer*, Insurance Mathematics, Diploma programme in Economics, Saint Petersburg State University.

Awards and Funding

Jul 2024	**Travel grant**, ISBA 2024 World Meeting
Sep 2022	**Travel grant**, ESOBE 2022 conference
Apr 2022	**Travel grant**, BNP 2022 networking event
1996 – 2001	**Merit-based scholarship** for undergraduate studies awarded by Saint Petersburg State University

Languages

English	fluent, native speaker proficiency (Cambridge CPE certificate)
German	fluent, full proficiency in writing, reading and speaking
French	advanced (C1), very confident in reading and writing, slightly less in speaking
Russian	native speaker

Computer Skills

- R, Python (NumPy, SciPy, sklearn, statsmodels, matplotlib), MATLAB, Stata, Mathematica, LaTeX

Service to Profession

Reviewer for journals: *Bayesian Analysis*, *Theoretical and Applied Climatology*

Memberships

Member of the International Society for Bayesian Analysis (ISBA) and the Royal Statistical Society (RSS).